

2026S1 PH1012: Physics A
Impulse, Momentum and Force

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Week 7
Serway Chap 9.1 – 9.5

“One way to keep momentum going is to have constantly greater goals..”

Michael Korda

Outline

Impulse: $\vec{J} = \int_{t_i}^{t_f} \vec{F} dt = \vec{F}_{avg} (t_f - t_i)$

Impulse-Momentum relation:

$$J = \int F dt = mv_{final} - mv_{initial}$$



Giancoli pg 163:

Now we analyze applying force over a period of time – Impulse
Later, we will analyze applying force over a displacement – Work Done



(a) Approach



(b) Collision



(c) If elastic



(d) If inelastic

Applying conservation of momentum:

$$m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f} = m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i}$$

Usually extra information required to solve problems

Special case: Elastic collision

$$\vec{v}_{1i} - \vec{v}_{2i} = -(\vec{v}_{1f} - \vec{v}_{2f})$$

Giancoli Fig 9.12:

Collision between two balls of same mass.

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. apply the impulse-momentum relation, whenever appropriate, to analyze real life scenarios involving objects with changing momentum
2. apply conservation of momentum, whenever appropriate, to analyze real life problems involving collisions, explosions in 1D and 2D
3. explain the origin of “propulsion forces” and solve problems involving propulsion forces
4. solve problems involving variable forces such as viscous forces and spring force

Impulse and Momentum Change (Force and Time)

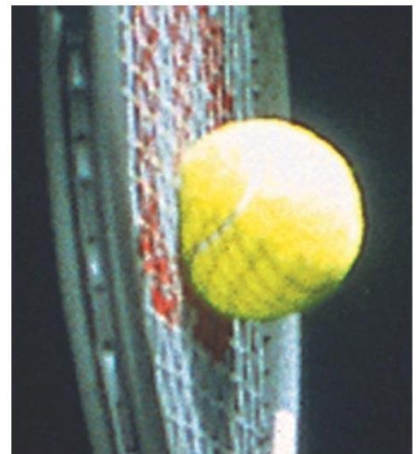
A quick recap:

Momentum p of an object is given by the product of its mass m with its velocity v
$$p = mv.$$

(Modified) Giancoli pg 216, Example 9-1

Calculate the momentum of a 57 g tennis ball moving horizontally at 65 m/s.

The tennis ball is struck by a racquet and it moves in the reverse direction at 70 m/s.
Calculate the new momentum.



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Giancoli pg 220 Fig 9.8:
Tennis racquet striking a ball.

Calculate the change in momentum of tennis ball.

This quantity, change in momentum, is also known as impulse. If we can control impulse, we can control the final momentum.

If the ball is in contact with the racquet for 4 ms, estimate the average force on the ball.



Energy and kinetic energy

Let us consider what happens when we apply a force F on an object of mass m over a distance d . So here we can conclude that the object will have an acceleration

$$a = F/m$$

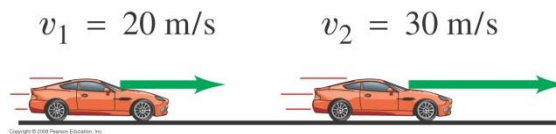
If the object has an initial velocity v_i and final velocity v_f , we can write

$$v_f^2 = v_i^2 + 2ad$$
$$v_f^2 - v_i^2 = 2 \frac{F}{m} d$$

Rearranging, we have

$$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = Fd$$

The quantity Fd is known as work done and the quantity $\frac{1}{2}mv^2$ is known as (translational) kinetic energy. We use the intuitive way to define energy at the moment – “the ability to do work”. A car (of mass 1200 kg) moving at 20 m/s accelerates to 30 m/s.



What is the kinetic energy of the car when it is travelling at 20 m/s?

$$K.E._1 = \frac{1}{2}mv^2 = \frac{1}{2} \times 1200 \text{ kg} \times \left(20 \frac{\text{m}}{\text{s}}\right)^2 = 240\,000 \text{ J}$$

What is the change in kinetic energy of the car? What is the work done by the engine to cause this change?

$$\Delta K.E._1 = \frac{1}{2}m(v_f^2 - v_i^2) = \frac{1}{2} \times 1200 \text{ kg} \times \left[\left(30 \frac{\text{m}}{\text{s}}\right)^2 - \left(20 \frac{\text{m}}{\text{s}}\right)^2 \right] = 300\,000 \text{ J}$$

If the distance for this change is 20 m, what is average force exerted by the engine to cause this change in velocity?

$$\Delta K.E. = \text{Work done by engine} = \text{Force} \times \text{displacement}$$
$$300\,000 \text{ J} = F \times 20 \text{ m} \Rightarrow F = 15\,000 \text{ N}$$

Impulse:

When a constant force $\vec{F}$ acts on an object, the impulse $\vec{J}$ is the force multiplied by the time interval $\Delta t = t_f - t_i$ during which the force acts.

$$\vec{J} = \vec{F}\Delta t$$

1. Note that impulse $\vec{J}$ is a vector quantity; its direction is the same as the force $\vec{F}$.
2. Units: Force x time (N.s or kg.m/s)
3. Area under F-t graph gives the impulse (or total change in momentum of an object over a given time interval).
4. As we have seen, this physical quantity is related to momentum change.

$$\vec{p}_f - \vec{p}_i = \vec{J}$$

Impulse and Momentum Change: How did we get $\vec{J} = \vec{F}\Delta t$?

As we have learnt previously for constant mass,

$$\vec{F} = m\vec{a} = m \frac{(\vec{v}_f - \vec{v}_i)}{\Delta t} = \frac{\vec{p}_f - \vec{p}_i}{\Delta t}$$

Thus,

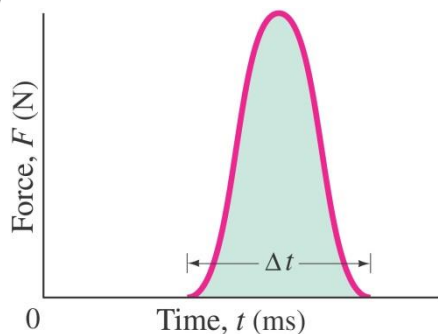
$$\vec{F}\Delta t = \vec{p}_f - \vec{p}_i = \vec{J}$$

Impulse-Momentum relation

When a constant force $\vec{F}$ acts on an object during a time interval $\Delta t = t_f - t_i$, the change in the object's momentum is equal to the impulse of the force acting on the object

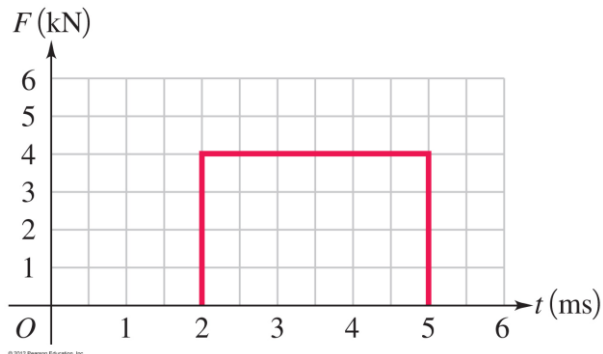
$$\vec{F}\Delta t = \vec{p}_f - \vec{p}_i = \vec{J}$$

Giancoli pg 220 Fig 9.9:
Force –time graph for a typical collision.



Impulse and Momentum Change:
Young (College Physics) pg 263 Prob 8.41

A 150 g baseball is hit towards the left by a bat. The magnitude of the force the bat exerts on the ball as a function of time is shown in the figure below:

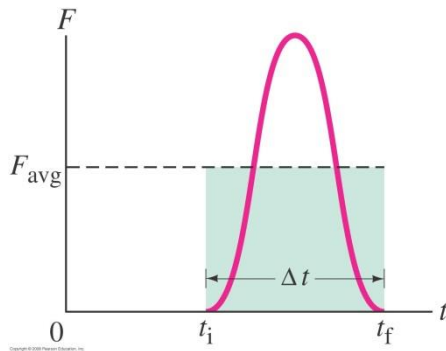


- Find the magnitude and direction of the impulse that the bat imparts to the ball.
- Find the ball's velocity just after it is hit by the bat if the ball is initially at rest.
- Find the ball's velocity just after it is hit by the bat if the ball is initially moving to the right at 30.0 m/s.

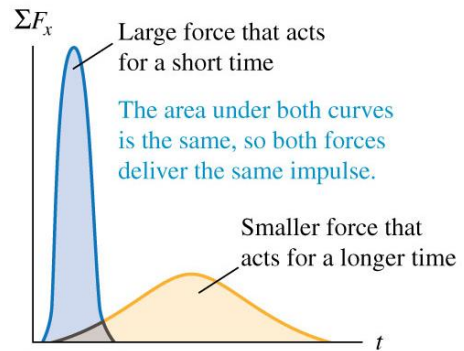


Generalization

The relation is also applicable to non-constant forces acting on a constant mass. In reality, when an object is struck, the force F acting on that object is like to vary in time t as shown below.



Giancoli pg 221 Fig 9.10:
Average force acting over time interval Δt .



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Young & Freedman pg 243 Fig 8.3b:
Same Impulse, different Δt .

The impulse is given by the area under the F-t graph:

$$\vec{J} = \int_{t_i}^{t_f} \vec{F} dt = \vec{F}_{avg} (t_f - t_i)$$

Application: Advice for the apprentice

The young apprentice is challenged to catch a piece of tofu thrown at him without breaking it. What would your advice to him be?



Other applications

1. Cat landing



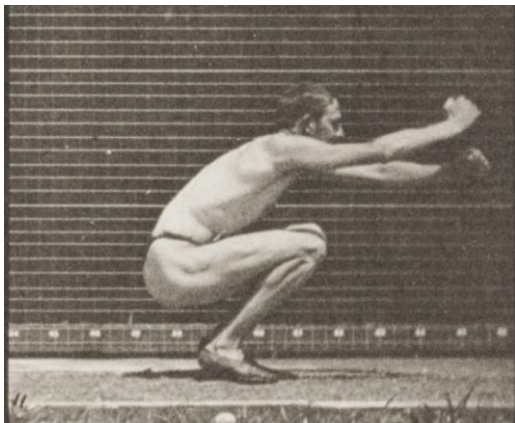
<https://pixabay.com/en/cat-jump-kitten-breed-cat-2219444/>

2. Airbag



<https://www.flickr.com/photos/thomashawk/6681328003>

3. Standing board jump (Landing)

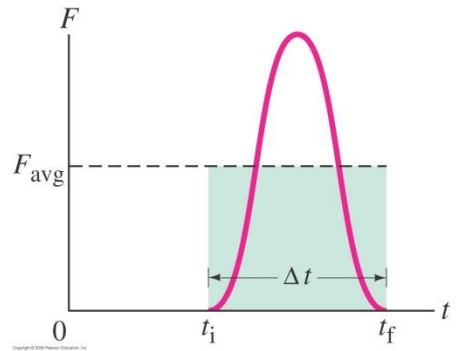


<http://digitallibrary.usc.edu/cdm/ref/collection/p15799coll58/id/5090>



Other applications 2

Golf / Baseball



Giancoli pg 221 Fig 9.10:
Average force acting over time interval Δt .

A different scenario / analysis



Impulse delivered
on object (ball) by
'external agent' (bat).

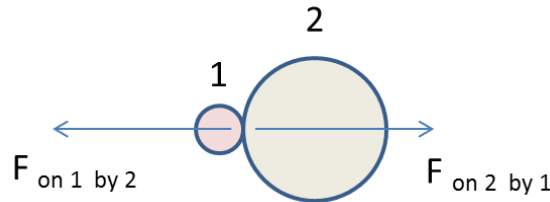


Impulse delivered on object
(car) by another object
(car). These two objects
form a closed system.

Conservation of Linear Momentum

Newton's third law + Impulse-Momentum relation

Consider two particles colliding into each other and exerting a force on each other throughout the duration $\Delta t = t_f - t_i$ of contact.



According to Newton's third law, $F_{12} = -F_{21}$. Applying the momentum-impulse relation, the change in momentum of particle 1 during the time interval of collision is (we simplify the analysis by assuming F is constant - the result can be generalized to cases where F is not constant in time),

$$\Delta \vec{p}_1 = \vec{F}_{12} \Delta t$$

Similarly for particle 2,

$$\Delta \vec{p}_2 = \vec{F}_{21} \Delta t$$

Combining the two equations we get

$$\Delta \vec{p}_1 + \Delta \vec{p}_2 = \vec{F}_{12} \Delta t + \vec{F}_{21} \Delta t = 0$$

since $F_{12} = -F_{21}$. Thus, the change in momentum of the system of the two particles is zero (when there are no other external forces acting).

Conservation of linear momentum states that the total linear momentum of an isolated system of particles is constant at all times.

Remarks:

1. We can write $\Delta \vec{p} = m\vec{v}_f - m\vec{v}_i$. Thus, for a two body collision,

$$(m_1 \vec{v}_{1f} - m_1 \vec{v}_{1i}) + (m_2 \vec{v}_{2f} - m_2 \vec{v}_{2i}) = 0$$

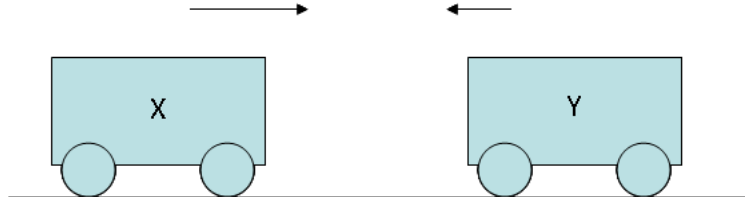
$$\mathbf{m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f} = m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i}}$$

which essentially says that the total momentum before collision is the same as the total momentum after collision.

2. This result applies to all collisions (even for many particle systems).
3. The same applies to a system of particles interacting at a distance (without touching).
4. It is important to identify the system correctly such that external forces are excluded before the conservation of momentum is applied.

**Conservation of Linear Momentum:
An Example: Two toy cars colliding 1 (Inelastic)**

The diagram below shows two toy cars, X (2 kg, 7 m/s right) and Y (3 kg, 3 m/s left), about to collide.



After the collision, the directions of motion of both toy cars are reversed and the magnitude of the momentum of X is then 2 N.s .

- a) What is then the corresponding momentum of Y?

- b) What is the velocity of Y?

- c) What is the change in total kinetic energy of the system?

Note that we have some information about the toy cars after the collision. In general, we need additional information on the nature of the collision in order to solve the problem.

Conservation of Linear Momentum: Elastic Collisions

There are two quantities conserved in elastic collisions: total momentum and kinetic energy. This is a special type of collision. Any collision where the kinetic energy is not conserved is known as an inelastic collision.

We consider a simple case with just two masses m_1 and m_2 moving at velocity u_1 and u_2 before the collision. After the collision, the two masses move off at v_1 and v_2 respectively. In many situations, we are interested to know the velocities v_1 and v_2 in terms of the variables m_1 , m_2 , u_1 and u_2 .

[Spot the maths errors in the derivation]

- Conservation of Linear Momentum (CoLM):

$$\begin{aligned} m_1 u_1 + m_2 u_2 &= m_1 v_1 + m_2 v_2 \\ m_1(u_1 - v_1) &= m_2(u_2 - v_2) \dots (1) \end{aligned}$$

- Conservation of Kinetic Energy (CoKE):

$$\begin{aligned} \frac{1}{2}m_1 u_1^2 + \frac{1}{2}m_2 u_2^2 &= \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 \\ m_1(u_1^2 - v_1^2) &= m_2(u_2^2 - v_2^2) \\ m_1(u_1 + v_1)(u_1 - v_1) &= m_2(u_2 + v_2)(u_2 - v_2) \dots (2) \end{aligned}$$

Dividing equation (2) by equation (1), we get

$$\begin{aligned} (u_1 + v_1) &= (u_2 + v_2) \\ u_1 - u_2 &= -(v_1 - v_2) \dots (3) \end{aligned}$$

Remarks:

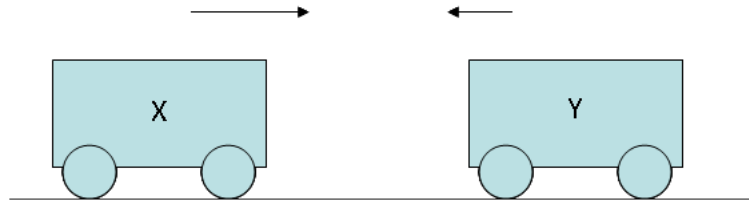
1. Equation (3) tells us that for elastic collisions:
Relative velocity of approach (before collision)
= Relative velocity of separation (after collision).

The relation is independent of mass. The masses m_1 , m_2 are not in Equation (3).

2. Solving equations (1) and (3) simultaneously gives us v_1 and v_2 .

An Example: Two toy cars colliding 2 (Completely Elastic)

The diagram below shows two toy cars, X (2 kg, 7 m/s right) and Y (3 kg, 3 m/s left), made from a special material different from the two previous examples, are about to collide.



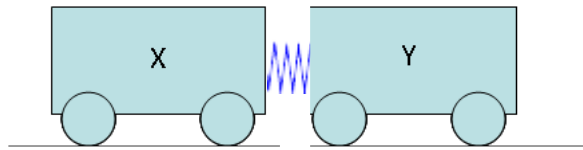
The collision is perfectly elastic. (Kinetic Energy is conserved)

a) What are the velocities of X and Y immediately after the collision?

b) What is the change in total kinetic energy of the system?

An Example: Two toy cars 'colliding' 4 ('Explosion')

The diagram below shows two toy cars, X (2 kg) and Y (3 kg) pushed together such that they compressed a spring. The two toy cars are at rest.



At time $t = 0\text{s}$, the two toy cars are released and X moves off at 5 m/s .

- a) What is the velocity of Y after the cars are released?

- b) What is the change in total kinetic energy of the system?



Conservation of Linear Momentum: Different Types of Collisions

1. **(Momentum)** For an isolated system of particles, the conservation of linear momentum states that the total linear momentum is constant at all times. Thus, in all types of collisions, the total linear momentum (immediately) before the collision = total linear momentum (immediately) after the collision.
2. **(Energy)** In addition, the total energy of a closed system is also constant for all collisions. (Here, we have to include energy such as heat, sound etc.)
3. **(Kinetic Energy)** For general (two body) collisions whereby kinetic energy of the system is lost to the surrounding as heat and sound, it can be very difficult to predict the final velocities from the masses and the initial velocities. We study some of these collisions where we can solve completely:
 - Completely Inelastic Collision: objects moving off with common velocity after collision.
 - Elastic Collision: The total kinetic energy of the system is conserved. (This is quite a special case.)
 - Explosion: A single mass disintegrating into two (or several) pieces.



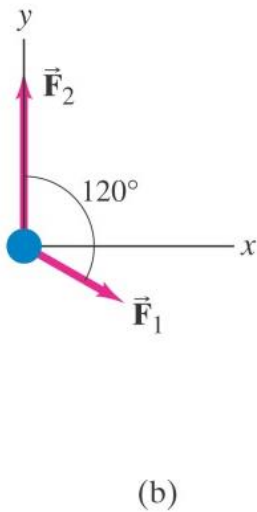
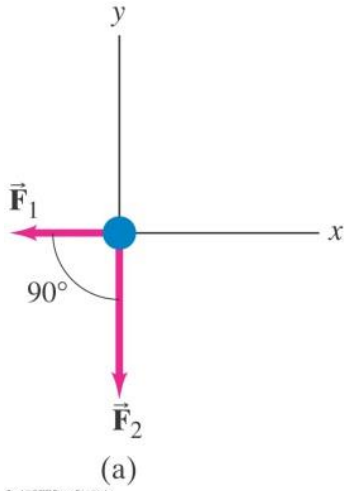
Some Reflections:

A basketball is released from a height above the ground. We note that the momentum of the ball is increasing, i.e. not conserved. Is there something wrong with the conservation of momentum?



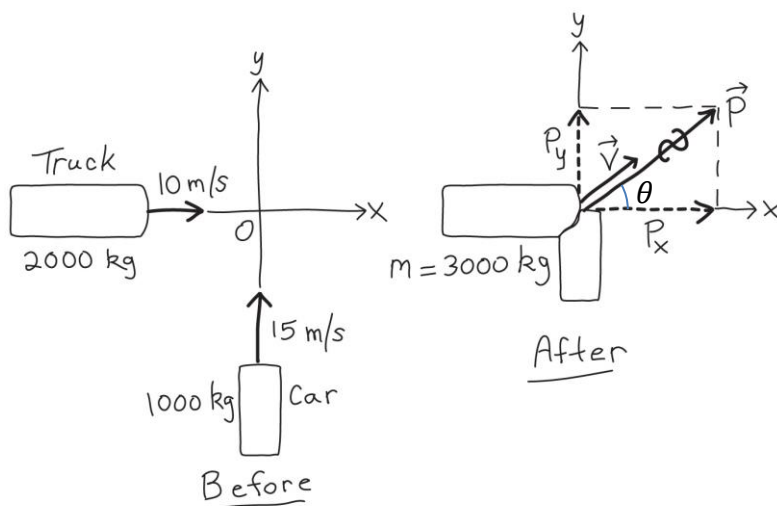
EXERCISE: Quick Revision on Vectors

The two forces $\vec{F}_1$ and $\vec{F}_2$ shown in the figures (top view) act on a 18.5-kg object on a frictionless table top. If $\vec{F}_1 = 10.2\text{ N}$ and $\vec{F}_2 = 16.0\text{ N}$, find the net force on the object and its acceleration for (a) and (b).



Conservation of Linear Momentum (2D): (Modified) Young & Freedman pg 254 Example 8.9

A small compact car with a mass of 1000 kg is travelling north along Scotts road with a speed of 15 m/s. At the intersection of Scotts and Orchard Road, it collides with a truck with a mass of 2000 kg that is travelling east on Orchard road at 10 m/s. The two vehicles become thoroughly tangled and moved away from the point of contact as one mass. Fortunately, all passengers were wearing seat belts and no one was injured. What is the velocity of the wreckage immediate after the impact? [For simplicity, treat each vehicle as a particle.]



Calculate the impulse acting on the truck during the collision.



2026S1 PH1012: Physics A
Propulsion

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Week 7 – R
Serway Chap 9.7 – 9.9

"Heavier than air flying machines are impossible."

Lord William Thomson Kelvin (1824-1907)

"I think there is a world market for maybe five computers."

Thomas Watson, chairman of IBM, 1943

Outline

$$F_{avg} = \frac{p_f - p_i}{\Delta t}$$

$$F_{avg} = v \frac{dm}{dt}$$



How does a helicopter hover over a point in air?



Some observations:

1. There is an upward force F_T on the helicopter that is equal to its weight.
2. This upward force F_T is related to the rotor blades because if it is not turning, the helicopter will fall.
3. So what is the reaction pair to the force F_T ? See the picture above for clues.

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A more visual approach

It actually involves the rotor blades pushing air downwards. But air is hard to see and visualize. Let us study something more concrete before returning to the helicopter.

Consider a man on ice skates on ice. He has an assistant next to him who is always moving at the same speed as him and passing him the balls to him to throw. If he starts from rest and throws an iron ball forward, we can expect him to acquire a momentum in the opposite direction (by conservation of momentum). If he keeps throwing iron balls in the same direction at a constant velocity (relative to him), we can expect his momentum p to progressively increase over time. Thus, we can work out the average force acting on him since

$$F_{avg} = \frac{p_f - p_i}{\Delta t}$$



If he throws the ball at a higher velocity, we expect his change in momentum to be larger. If he throws many iron balls per unit time, his gain in momentum will also be large.

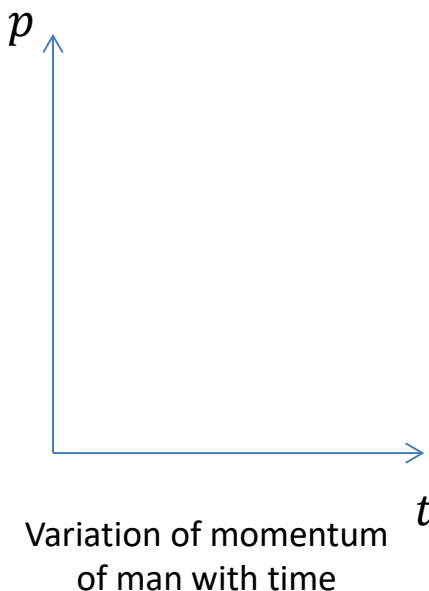
Thus, the two important factors that determines the magnitude of this average force, F_{avg} :

- 1.
- 2.

Hence, we can write

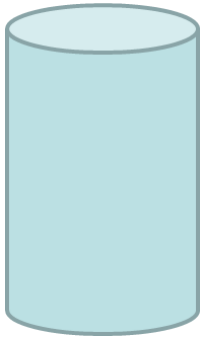
$$F_{avg} = v \frac{dm}{dt}$$

where v is the velocity of the propelled fluid (air in the case of the helicopter) and $\frac{dm}{dt}$ is the mass flow rate of the fluid.



Calculation: helicopter hovering over a point in air.

Returning to our helicopter question. Instead of throwing iron balls, the rotor blades of the helicopter are 'throwing' air downwards to lift itself upwards. (I find it hard to imagine this too but let's see the numbers.) Consider a helicopter with rotors blades of length 7 m and can produce a downstream airspeed of 20 m/s. Assuming that the density of air is 1.21 kg m^{-3} , compute the upward force F_T produced by the rotor blades.



Volume of air swept out by
rotor blades every second.

Example: [G9.86] During a Chicago storm, winds can whip horizontally at speeds of 120km/h. If the air strikes a person at the rate of 45 kg/s per square meter and is brought to rest, calculate the force of the wind on a person. Assume the person is 1.60 m high and 0.50 m wide.

Other Daily Examples

[Object 'pushing' surrounding fluid]

1. Swimming - (why do you have to keep your fingers together to swim effectively?)



2. Rowing



[Surrounding fluid 'pushing' against object]

3. Air Resistance

Which car is good for racing? $F_D = \frac{1}{2}C_D\rho Av^2$



<https://www.videoblocks.com/video/air-flow-around-pickup-truck-in-wind-tunnel-with-slow-motion-smoke-following-form-of-body--profile-side-view-3d-animation-simulation--art-bdanhx2ngj0y0hrz4>



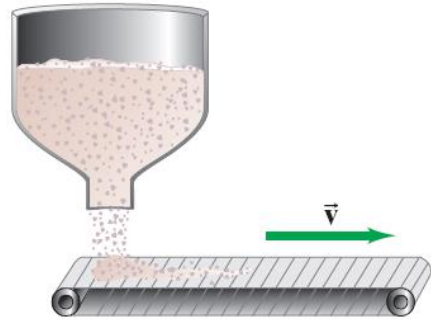
<https://jaimeirastorza.wordpress.com/2014/05/25/notes-on-car-aerodynamics-for-beginners/>



<https://www.pinterest.com/acondemella do/aerodynamics/?lp=true>

Giancoli pg 237 Example 9-19

You are designing a conveyor system for a gravel yard. A hopper drops gravel at a rate of 75.0 kg/s onto a conveyor belt that moves at a constant speed $v = 2.20 \text{ m/s}$. Determine the additional force (over and above internal friction) needed to keep the conveyor belt moving as gravel falls on it.



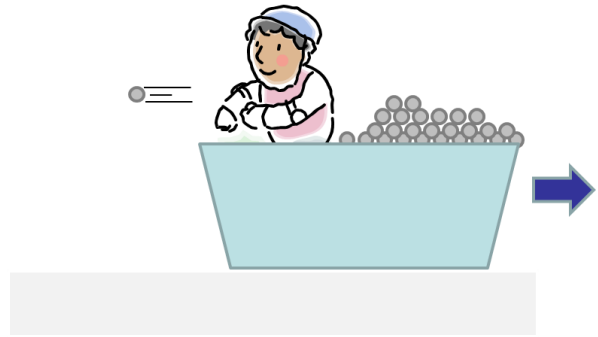
In 1s ,

Total horizontal momentum change of gravel =

Systems with variable mass [Non-examinable]

Rockets in Space ['No more surrounding fluid']

We have learnt how to propel ourselves forward. However, if we want to go to outer space, we have a problem - there is no air in space and there is nothing else to 'push' backwards like in swimming. Therefore, one needs to bring along some mass (such as liquified air) to propel the rocket



We use our man throwing iron balls again to help in our analysis. Now he is in a sledge on ice filled with iron balls of mass. We will ignore friction. As before, when he throws the iron balls at a constant rate in one direction always with the same velocity (relative to him), he gains momentum in the opposite direction. However, there is one complication in the analysis: the mass of the sledge+man is not constant. Therefore, acceleration will not be constant and velocity will not increase at a constant rate.

Systems with variable mass

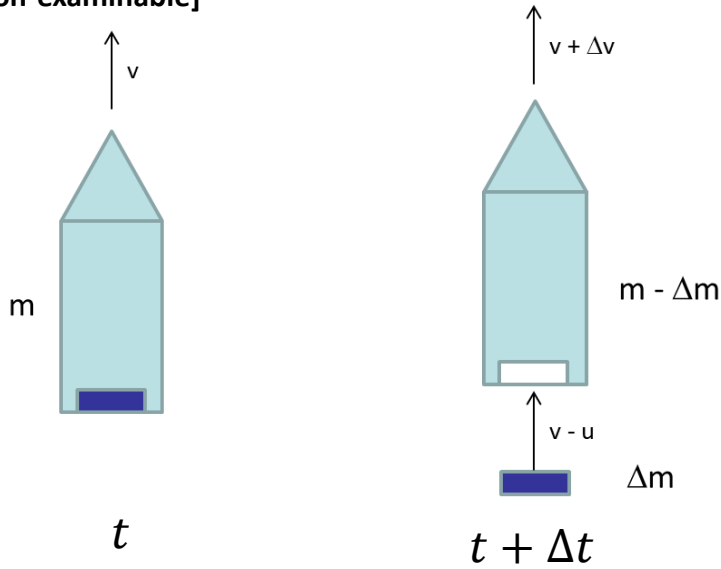
We now consider a system whereby the mass varies with time. In the most general case, we consider Newton's second law:

$$\Sigma \vec{F}_{ext} = \frac{d\vec{P}}{dt}$$

where $\vec{F}_{ext}$ is the net external force acting on the system and $\vec{P}$ is the total momentum. The change in momentum of the system is given by

$$\Delta \vec{P} = \vec{P}_{final} - \vec{P}_{initial}.$$

Rockets [Non-examinable]



We will derive the change in momentum in the most general case (in the presence of a non-zero net external force):

$$\Delta \vec{P} = \vec{P}_{final} - \vec{P}_{initial}$$

[dropping $\vec{}$ from this point for simplicity]

$$\Delta p = (m - \Delta m)(v + \Delta v) + \Delta m(v - u) - mv$$

$$\Delta p = m\Delta v - \Delta m\Delta v - \Delta m \times u$$

If the change in mass Δm and velocity Δv happens over a very small period of time Δt , we divide throughout by Δt and take the limits $\lim_{\Delta t \rightarrow 0}$,

$$\Delta p = m \frac{\Delta v}{\Delta t} - \Delta m \frac{\Delta v}{\Delta t} - \Delta m \frac{\Delta v}{\Delta t} \times u$$

$$F_{ext} = \frac{dp}{dt} = m \frac{dv}{dt} - 0 - u \frac{dm}{dt}$$

If the rocket is far away from any planets etc, the net external force is zero [the rocket and its exhaust form a closed system]. In the above equation $u \frac{dm}{dt}$ refers to the force on the exhaust, the reaction pair on the rocket (in the opposite direction) should pick up negative sign. So rearranging the equation, we get

$$m \frac{dv}{dt} = \cancel{F_{ext}} - u \frac{dm}{dt}$$

commonly known as the 0 in free space
rocket equation. We note here that the mass of the rocket m is not constant.

Rockets in Space 3 [Non-examinable]

The rocket equation (in free space)

$$m \frac{dv}{dt} = -u \frac{dm}{dt}$$

Solving the equation, we see how the mass of the rocket depends on time:

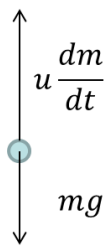
$$\begin{aligned} dv &= -u \frac{dm}{m} \\ \int_{v_i}^{v_f} dv &= -u \int_{m_i}^{m_f} \frac{dm}{m} \\ v_f - v_i &= -u [\ln(m_f) - \ln(m_i)] \\ v_f - v_i &= u \left[\ln\left(\frac{m_i}{m_f}\right) \right] \end{aligned}$$

Remarks:

1. The initial mass m_i is always larger than the final mass m_f because fuel is burnt.
2. If we have a large final velocity v_f when we burn out all the fuel, we need to eject the fuel at a large velocity u and design the rocket such that a large percentage of the rocket's mass is to be burnt out, i.e. large $\frac{m_i}{m_f}$ ratio.
3. Note that while the gain in velocity is always the same for the same $u \ln\left(\frac{m_i}{m_f}\right)$ but the gain in kinetic energy is also dependent on the initial velocity v_i .

Rockets from Earth [Non-examinable]

Now if we are launching the rocket from Earth, there is an external force acting on the rocket -- its weight mg . So now, we have to modify the rocket equation, (launching from Earth)



$$m \frac{dv}{dt} = mg - u \frac{dm}{dt}$$

(Launching from Earth) The derivation for the launching for Earth case is similar (so we will skip it) but the solution only requires a minor adjustment:

$$v_f - v_i = u \left[\ln\left(\frac{m_i}{m_f}\right) \right] - gt$$

Giancoli pg 238 Example 9-20: Rocket propulsion. [Non-examinable]

A fully fueled rocket has a mass of 21,000 kg, of which 15,000 kg is fuel. The burned fuel is spewed out the rear at a rate of 190 kg/s with a speed of 2800 m/s relative to the rocket. If the rocket is fired vertically upward calculate: (a) the thrust of the rocket; (b) the net force on the rocket at blastoff, and just before burnout (when all the fuel has been used up); (c) the rocket's velocity as a function of time, and (d) its final velocity at burnout. Ignore air resistance and assume the acceleration due to gravity is constant at $g = 9.80 \text{ m/s}^2$.

a) $F_{thrust} = v_{rel} \frac{dM}{dt} = \left(-2800 \frac{\text{m}}{\text{s}}\right) \left(-190 \frac{\text{kg}}{\text{s}}\right) = 5.3 \times 10^5 \text{ N}.$

b) Weight of the rocket at blastoff,

$$W_I = M_I g = (2.4 \times 10^4 \text{ kg}) \left(9.80 \frac{\text{m}}{\text{s}^2}\right) = 2.1 \times 10^5 \text{ N}$$

Weight of the rocket at burnt out,

$$W_F = M_F g = (6.0 \times 10^3 \text{ kg}) \left(9.80 \frac{\text{m}}{\text{s}^2}\right) = 5.9 \times 10^4 \text{ N}$$

At Blast off: $F_{net, I} = F_{thrust} - W_I = 3.2 \times 10^5 \text{ N}$

At Burnt out: $F_{net, F} = F_{thrust} - W_F = 4.7 \times 10^5 \text{ N}$

c) Referring Rockets from Earth on pg 9,

$$M \frac{dv}{dt} = Mg - u \frac{dM}{dt}$$

Solving, we get

$$v_f - v_i = u \left[\ln \left(\frac{m_i}{m_f} \right) \right] - gt$$

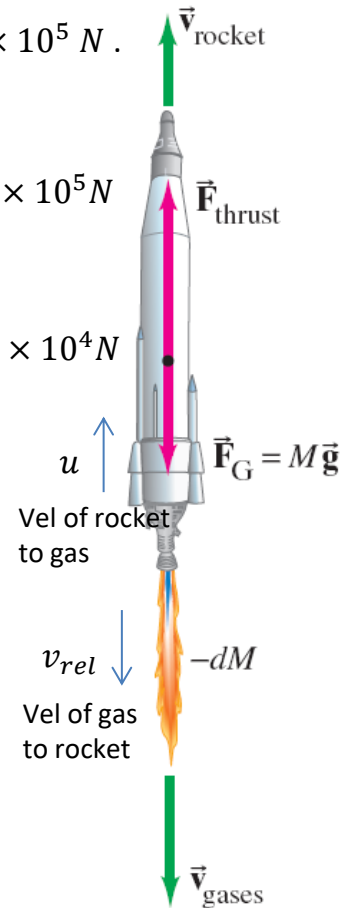
[Note, $u = -v_{rel}$.]

d) Time required for burn out 15 000kg at 190kg/s,

$$t = \frac{15000}{190} = 79 \text{ s}.$$

If we take $v_i = 0$, then

$$v = -9.8 \frac{\text{m}}{\text{s}^2} \times 79 \text{ s} + (2800 \frac{\text{m}}{\text{s}}) (\ln 6000/15000) = 2700 \text{ m/s}$$





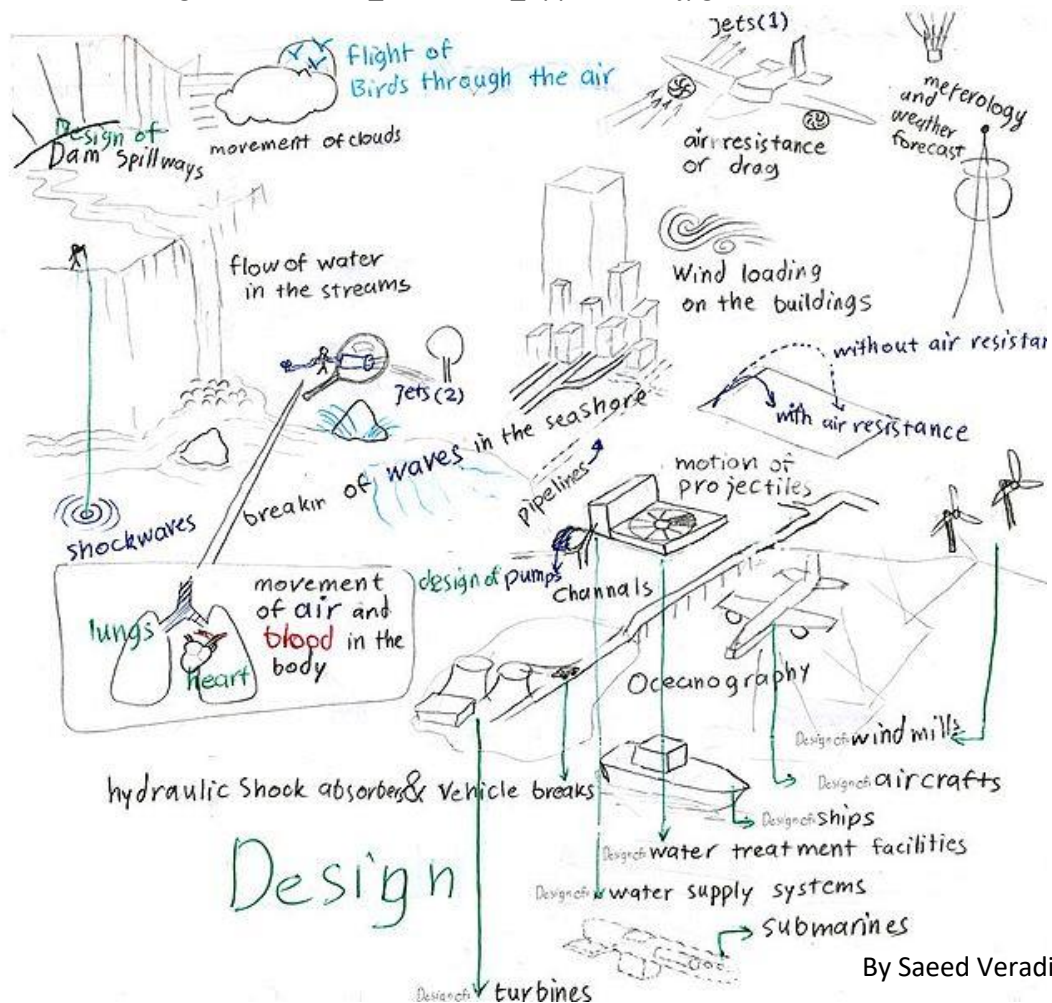
"I've been noticing gravity since I was very young."

- Cameron Diaz

"One story Einstein liked to tell about his childhood was of a wonder he saw when he was four or five years old: a magnetic compass. The needle's invariable northward swing, guided by an invisible force, profoundly impressed the child. The compass convinced him that there had to be "something behind things, something deeply hidden."

About Albert Einstein

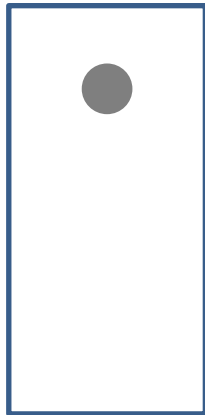
https://commons.wikimedia.org/wiki/File:Fluid_Mechanics_Applications.jpg



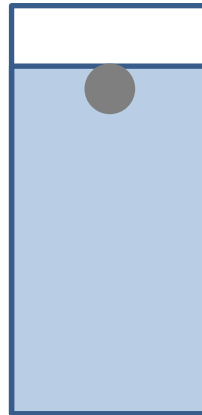
Viscous Forces

Which will hit the bottom first?

Consider a steel ball released in a vacuum and one released into a jar of detergent from the same height. Which will hit the bottom first?



In vacuum



In detergent

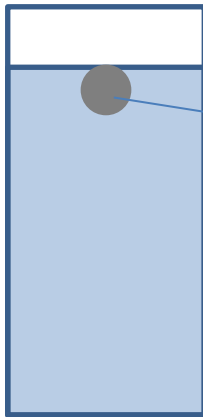
For an object moving through a liquid at relatively low speeds, the drag force F_v will typically be proportional velocity, v .

$$\vec{F}_v = -b\vec{v}$$

For an object moving through a air, the drag force F_D will typically be proportional velocity, v^2 . How the drag force is dependent on velocity is usually determined by the “Reynolds number” (in fluid mechanics).

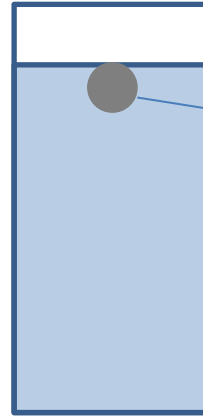
Drag Forces: Which will hit the bottom first? Part 2

$$\vec{F}_v = -b\vec{v}$$



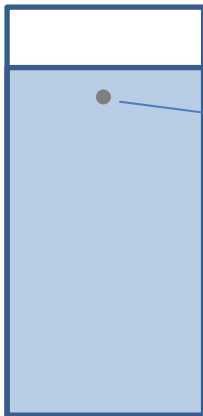
Steel ball

In detergent



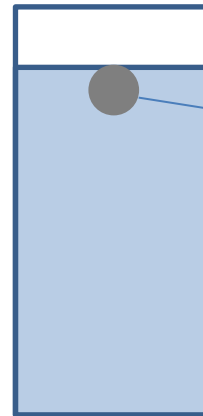
Marble of same size

In detergent



A) Small steel ball

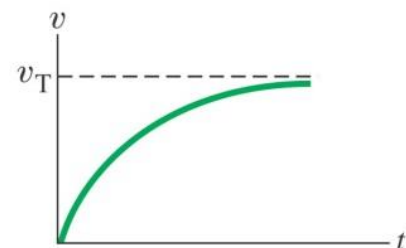
In detergent



B) Marble with same mass as small steel ball.

In detergent

Terminal Velocity



(b)

Drag Forces (Force Proportional to velocity)

Giancoli pg 130 Example 5-17:

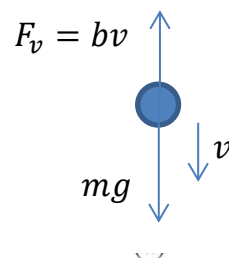
Determine the velocity as a function of time for an object falling vertically from rest when there is resistive force linearly proportional to v .

Applying Newton's 2nd law,

$$\vec{F}_v = -b\vec{v}$$

$$mg - bv = ma$$
$$\frac{mg}{b} - v = \frac{m}{b} \frac{dv}{dt}$$

$$v - \frac{mg}{b} = -\frac{m}{b} \frac{dv}{dt}$$



Note that $v = 0$ when $t = 0$

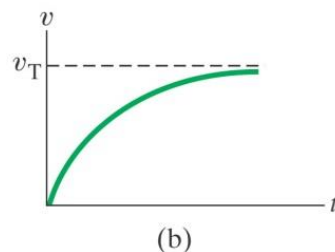
$$\int_0^v \frac{dv}{v - \frac{mg}{b}} = -\int_0^t \frac{b}{m} dt$$

which gives

$$\ln\left(v - \frac{mg}{b}\right) - \ln\left(0 - \frac{mg}{b}\right) = -\frac{b}{m}t \Rightarrow \ln\left(\frac{v - \frac{mg}{b}}{-mg/b}\right) = -\frac{b}{m}t$$

Simplifying, we get

$$v - \frac{mg}{b} = -\frac{mg}{b} e^{-\frac{b}{m}t} \Rightarrow v = \frac{mg}{b} (1 - e^{-\frac{b}{m}t})$$



As $t \rightarrow \infty$, $v \rightarrow v_T = \frac{mg}{b}$. v_T is the terminal velocity.

Giancoli pg 135 Problem 5-66

The terminal velocity of a $3 \times 10^{-5} \text{ kg}$ raindrop is about 9 m/s . Assuming a drag force $F_D = -bv$, determine (a) the value of the constant b and (b) the time required for such a drop, starting from rest, to reach 63% of terminal velocity.

At terminal velocity:

a) $mg - bv_T = ma = 0$

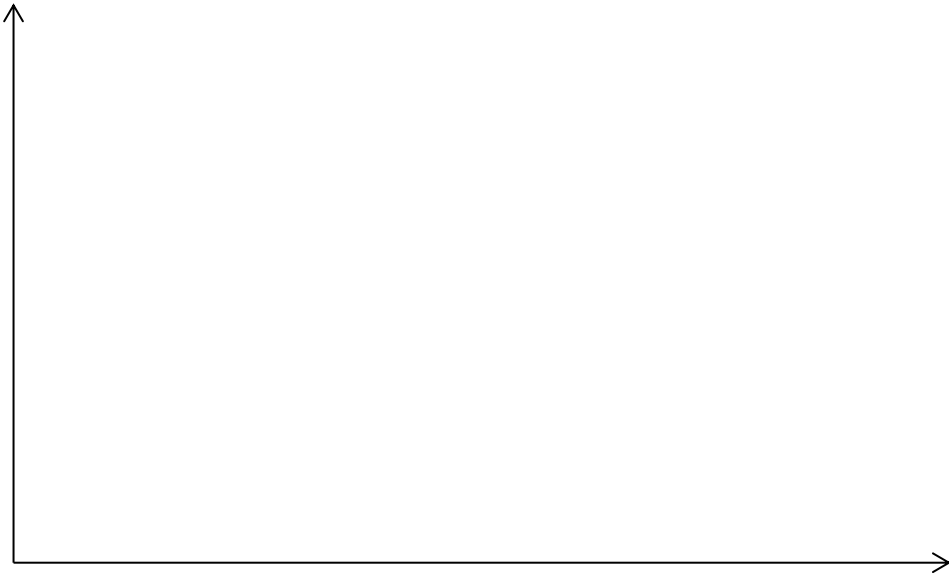
$$b =$$

b)

Just for fun: Scene from The Golden Eye

HOSY@ntu.edu.sg

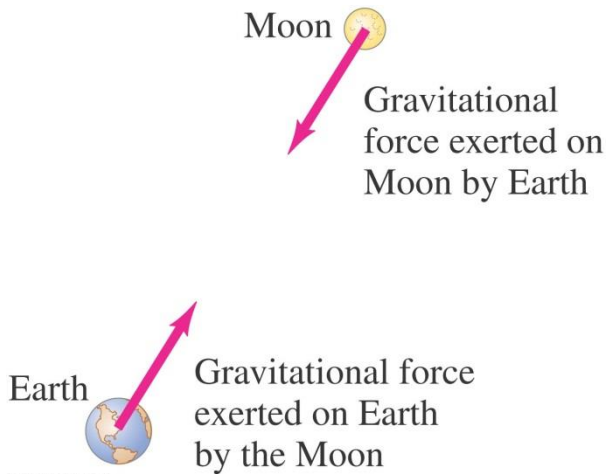
Think Critically: The plane falls off the cliff before James. Is it possible for James to catch up with the plane?



Examples of other (position/velocity dependent) forces

Gravitational force

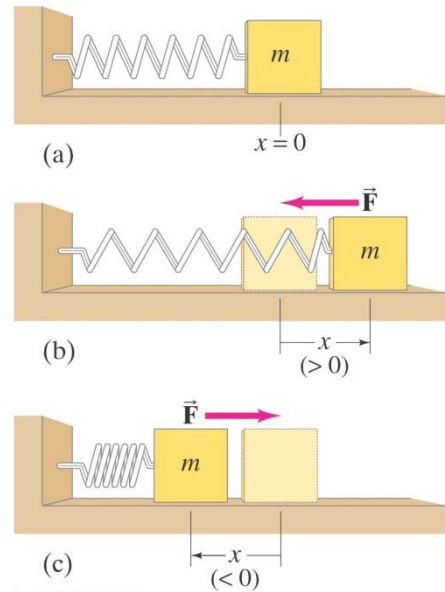
$$F = G \frac{m_1 m_2}{r^2}$$



Giancoli Fig 6.2

Spring force

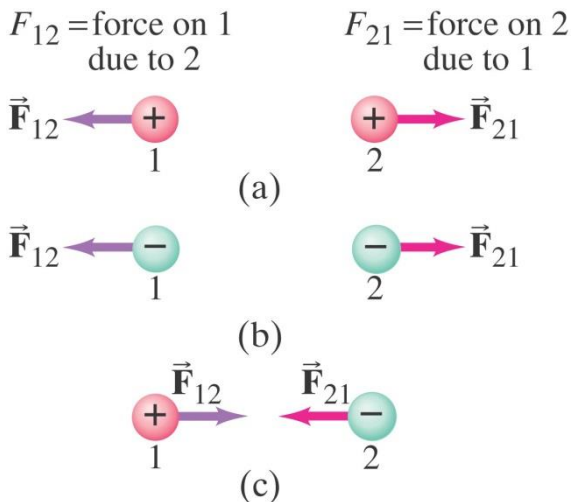
$$\vec{F} = -k\vec{x}$$



Giancoli Fig 14.1

Electric force

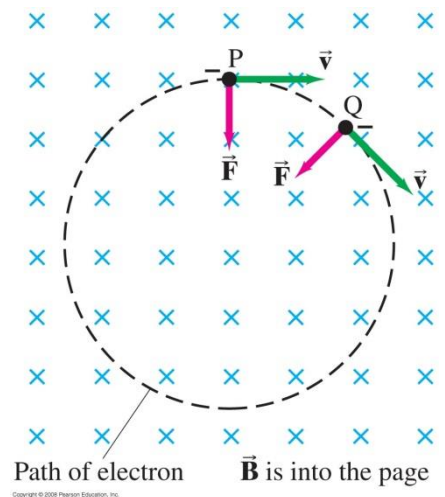
$$F = \frac{q_1 q_2}{4\pi\epsilon r^2}$$



Giancoli Fig 21.25

Magnetic force

$$F = qvB\sin\theta$$



Giancoli Fig 27.17